FoliaTrace AI

A browser tool that turns supplement labels and medicine descriptions into a source-linked pharmacist brief.

Problem, purpose and users

Older adults and caregivers may know a medicine only as "my cholesterol pill," while supplement labels contain ambiguous plant names, blends and repeated ingredients. FoliaTrace AI brings these details together before a pharmacy consultation. Users prepare the brief; pharmacists review the medicine-group suggestions, ingredient questions, NIH quotations and relevant studies.

How to use the working prototype

1. Open **Check a bottle**: search NIH product labels, paste ingredients or upload a label photo and review the extracted text. **2.** Add medicine names, brands or descriptions, the supplement goal and relevant circumstances. **3.** Select **Build my brief**; inspect the matched source passages and ranked studies. **4.** Use **Hand to pharmacist (QR)**, Print / PDF or Download; select which personal details to include.

AI technology and implementation

Medicine language: exact-name, brand and typo matching runs first. **all-MiniLM-L6-v2 sentence embeddings** compare unresolved descriptions with medicine-group prototypes using cosine similarity. The accepted suggestion remains beside the original wording for confirmation.

Research: BM25 retrieval selects up to 20 candidates from 423 open-access abstracts. The **ms-marco-MiniLM-L6-v2 cross-encoder** reranks them for each herb and user goal; up to two leads per herb clear a development-calibrated relevance threshold. Leads display the study's population, preparation and comparator excerpts.

Evidence and delivery: ingredient/medicine rules select from 74 verbatim NIH passages covering 25 herbs. Optional **Tesseract LSTM OCR** reads photos. ONNX models run in browser workers; QR handoff encodes the brief in a compressed URL fragment.

Four completed user workflows - input to observed result

Re-run on the public site on 13 November 2026. Profiles and photo label are fictional; the fourth workflow fetches a live NIH product record.

Scenario / input	Observed product output
Mei, 72 Warfarin + "her water pill"; memory/joint blend with ginkgo, turmeric, garlic and ginseng.	AI suggests a blood-pressure medicine group. Ginkgo/garlic medicine cards quote NIH passages for warfarin. Ginseng identity and blend-dose questions remain visible. Research separates ginkgo for memory from turmeric for joint pain.
Sam, 29 Sertraline + birth control pill; St. John's wort, valerian and chamomile; mood/stress goal.	Two medicine-specific NIH cards identify St. John's wort concerns for sertraline and birth control. Chamomile is presented as a species-identification question with candidate plants.
Ray, 61 Metformin + "my cholesterol pill" + lisinopril; two products containing red yeast rice, goldenseal and green tea.	AI maps the description to a statin group, adding the red yeast rice card missed by name matching. Goldenseal/metformin also appears. Green tea is flagged across both bottles; unrelated goldenseal research is withheld.
Own-bottle workflow Live NIH ginkgo label + photo; Coumadin, "zolpidm," "my blood pressure pill," ibuprofen; sleep goal.	Coumadin resolves to warfarin; "zolpidm" to zolpidem. AI suggests a blood-pressure medicine group. Ginkgo/Coumadin and valerian/zolpidem cards appear, with duplicate ginkgo and unresolved ibuprofen. A fresh browser opens the same brief via QR link; the medicine-excluded download omits all four names.

Measured AI and label-recognition results

Medicine language: 37/39 synthetic medicine descriptions assigned correctly, 0 assigned to the wrong group and 2 left unassigned; all 15 negative/other phrases left unassigned. **Label matching:** 789 held-out NIH records, 96.3% exact-species precision and 99.3% genus-or-better precision.

Try the product - choose Mei, Sam or Ray to reproduce the first three workflows.
Source + evidence - scenario inputs/results, evaluation datasets, model details and AI disclosure.